



TRAINING PROGRAMME

Machine Learning Engineer on AWS

Prepare data, train, deploy, automate and monitor models with SageMaker AI

45 HOURS

FOR Data scientists and engineers who put machine learning models into production on AWS

PREPARES FOR AWS Certified Machine Learning Engineer – Associate (MLA-C01)



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Overview

This programme teaches participants to build and operate machine learning systems on Amazon SageMaker AI and prepares the AWS Certified Machine Learning Engineer – Associate exam (MLA-C01). They prepare and label data, engineer features and check bias, train and tune models, deploy them to endpoints and batch transform, automate the lifecycle with SageMaker Pipelines and the Model Registry, monitor drift and cost, secure the workloads, and bring generative AI into production with Bedrock. Exam practice closes the content, and the programme ends with a production ML system built and presented by each team.

At a glance

#	PHASE	FOCUS	HOURS
01	ML Infrastructure & Data Preparation	SageMaker AI infrastructure, ingestion, Data Wrangler, Feature Store, bias checks	9 h
02	Model Development	Built-in algorithms, script mode, distributed training, tuning, evaluation	6 h
03	Deployment & MLOps	Endpoints, autoscaling, SageMaker Pipelines, Model Registry, CI/CD, monitoring	9 h
04	Security, Generative AI & Exam Preparation	Least privilege, VPC isolation, encryption, Bedrock in production, MLA-C01	9 h
05	Final Project	A production ML system on SageMaker AI, presented to the group	6 h
	Buffer	Catch-up, revision, extra lab time	6 h
	Total		45 h

Key facts

AUDIENCE	Data scientists and engineers who put machine learning models into production on AWS
PREREQUISITES	Machine learning (course: Machine Learning & Deep Learning) and AWS AI services (course: AWS AI Practitioner & Generative AI), with core AWS services (course: AWS Cloud Practitioner), or equivalent. ML theory is not taught again
DURATION	45 h: 39 h of chapters + 6 h buffer
FORMAT	Short lectures followed by guided labs in SageMaker Studio; one model carried through every phase; exam practice
LAB ENVIRONMENT	One laptop per participant with a browser; one AWS sandbox account per team (SageMaker AI domain, GPU quota, budget alert), provided by the organiser; endpoints deleted and Canvas sessions logged out at the end of each lab to control cost
ASSESSMENT	A lab validated at the end of each phase, practice exam questions and a final project presented to the group; certificate of completion for participants who complete the labs and the project
OUTCOME	Participants can build an automated pipeline from data preparation to a monitored endpoint on SageMaker AI, secure it, and sit the MLA-C01 exam

PHASE 01

9 H

ML Infrastructure & Data Preparation

Participants set up the infrastructure every later phase uses: a SageMaker AI domain, execution roles, S3, GPU instances and FSx. They then ingest and prepare data with Glue and Data Wrangler, store features in Feature Store and check the data for bias with Clarify before training. Validated lab: a feature group in Feature Store built from raw data, with a Clarify pre-training bias report.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	ML Infrastructure	SageMaker AI domains, IAM execution roles, S3, GPU instances, FSx	Lecture + Lab	3 h
1.2	Data Ingestion & Preparation	Glue and Glue DataBrew, Data Wrangler in SageMaker Canvas, streaming ingestion (Kinesis), data formats (Parquet, CSV, RecordIO), data quality, labelling with Ground Truth	Lecture + Lab	3 h
1.3	Feature Engineering & Bias	Feature Store, transformations, class imbalance, PII handling, Clarify pre-training bias	Lecture + Lab	3 h

PHASE 02

6 H

Model Development

Participants train models with built-in algorithms and their own framework code in script mode, scale training across instances, track experiments, then tune hyperparameters automatically and evaluate and explain the results. Validated lab: a tuning job whose best model beats the baseline on the evaluation metric, with a Clarify explainability report.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Training	Choosing AI services, Bedrock, JumpStart or custom training; built-in algorithms, script mode, managed spot training, distributed training, SageMaker Debugger, experiments with managed MLflow	Lecture + Lab	3 h
2.2	Tuning & Evaluation	Automatic model tuning, evaluation metrics, Clarify	Lecture + Lab	3 h

PHASE 03

9 H

Deployment & MLOps

Participants choose the right inference option, real-time, serverless and asynchronous endpoints and batch transform, with autoscaling, automate the lifecycle with SageMaker Pipelines, the Model Registry and CI/CD, then watch production with Model Monitor, drift detection, CloudWatch and cost controls. Validated lab: a pipeline that registers a model, deploys it after approval and raises an alert when the input data drifts.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Deployment	Real-time, serverless and asynchronous endpoints, batch transform, multi-model endpoints, Inference Recommender, custom containers in ECR, autoscaling	Lecture + Lab	3 h
3.2	MLOps	SageMaker Pipelines, Model Registry, CloudFormation and CDK, EventBridge, CodePipeline; blue/green, canary and shadow deployments	Lecture + Lab	3 h
3.3	Monitoring & Cost	Model Monitor (data quality, model quality, bias and feature-attribution drift), A/B testing, CloudWatch, CloudTrail, Cost Explorer, Savings Plans and Spot	Lecture + Lab	3 h

PHASE 04

9 H

Security, Generative AI & Exam Preparation

Participants secure ML workloads with least-privilege IAM, VPC isolation and encryption, bring generative AI into production with Bedrock fine-tuning, Knowledge Bases and Agents, then practise MLA-C01 questions. Validated lab: a training job and endpoint running without internet access inside a VPC, with encrypted data and model artifacts.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	Security for ML	IAM least privilege, VPC isolation, encryption, compliance	Lecture + Lab	3 h
4.2	Generative AI in Production	Bedrock customisation (a small fine-tuning job, on-demand inference), Knowledge Bases, Agents	Lecture + Lab	3 h
4.3	MLA-C01 Preparation	Practice questions and review	Lecture + Workshop	3 h

PHASE 05

6 H

Final Project

Teams of two or three build a production ML system for a provided use case, reusing the phase labs, then present it as they would to a head of data.

#	CHAPTER	CONTENT	FORMAT	HOURS
5.1	Brief & Build	Read the brief, build the pipeline from data preparation to the Model Registry	Project	3 h
5.2	Deploy & Monitor	Automated deployment, drift alerts, a Bedrock-backed feature; architecture diagram	Project	1.5 h
5.3	Demo & Review	Live run of the pipeline and a drift alert, cleanup; questions and feedback	Project	1.5 h

FINAL PROJECT

A Production ML System on SageMaker AI

A SageMaker pipeline from data preparation to the Model Registry, automated deployment to an endpoint, drift alerts in production, and a feature backed by Amazon Bedrock.

STACK AWS: SageMaker AI (Studio, Pipelines, Model Registry, Model Monitor, Clarify, Feature Store), Glue, S3, CloudWatch, Amazon Bedrock

DELIVERABLE Pipeline code, architecture diagram, monitoring dashboard, live demo